



SIGATAPU SHALEM RAJU

✉ shalemofficial1@gmail.com | [in](#) Shalem

☎ +91 6282986043 | [G](#) Shalem-2024



SKILLS

Generative AI & Agents	RAG, LangGraph, Agentic Workflows, Function Calling, Vector Databases (FAISS), State Management, Prompt Engineering.
LLM Engineering	Transformer Architecture, Hugging Face, Tokenization, Context Window Optimization, Semantic Search.
Model Tuning & Ops	Fine-Tuning, PEFT (LoRA/QLoRA), Quantization, Knowledge Distillation, Model Pruning, Adapter Layers, GPU Utilization.
Machine Learning	XGBoost, Scikit-learn, Feature Engineering, Hypothesis Testing, A/B Testing, Model Evaluation.
MLOps & Infrastructure	Docker, Apache Airflow, MLflow, FastAPI, PySpark, Databricks, SQL, AWS EC2, AWS S3.

WORK EXPERIENCE

Digital Twin Engineer
TVS Motor Company — R&D

Jan 2024 – Present
Bengaluru, KA

Project: Multi-Agent HIL Test Controller for Autonomous Test Case Execution

[Demo Repo](#)

- **Multi-Agent LLM Architecture:** Architected a multi-agent system (**Planner** and **ReAct Executor**) using **GPT-4o** to autonomously translate natural language instructions into executable **Hardware-in-the-Loop (HIL)** test sequences.
- Developed a matching engine utilizing OpenAI vector embeddings (**text-embedding-3-small**) to accurately map ambiguous user inputs to over 100+ complex **CAN bus signals** and Engine Management System parameters via cosine similarity.
- **Human-in-the-Loop UI:** Built a stateful **Streamlit** interface to facilitate human review, enabling engineers to dynamically modify AI-generated **JSON** test plans before dispatching them to **dSPACE AutomationDesk** via COM interfaces (**win32com**).

Project: Fine-Tuning Large Language Models for Proprietary Code Generation

[Repo](#)

- **Fine-Tuning & Optimization:** Leveraged the **Axolotl** framework to fine-tune **Qwen-2.5-72B** on **H200 GPUs**. Implemented **QLoRA** (Quantized Low-Rank Adaptation) to optimize VRAM usage (~40GB) while maintaining high-fidelity code generation.
- **Inference Engineering:** Built a production-grade inference script using **vLLM** and **PyTorch**. Engineered a ranked keyword filtering algorithm to dynamically select relevant library objects, solving critical context window bottlenecks.
- **Structured Output Generation:** Solved the challenge of generating strict, proprietary XML schemas from unstructured text by training the model on “Precondition/Action/Postcondition” data pairs.
- **Infrastructure Management:** Deployed and managed ephemeral GPU clusters on **RunPod** (A100/H100), utilizing persistent volume storage for model weight management and reproducible environment setups.

Project: Digital Twin Framework for EV Battery Health Analysis

[Repo](#)

- **Architected** a Digital Twin framework on Databricks to analyze battery degradation across **1.6M+ EVs**, transitioning the team from local Jupyter workflows to a scalable **PySpark/ADLS** architecture.

- **Engineered automated ELT pipelines** using **Databricks Jobs** to process 2.5 years of telematics data (219 parameters), reducing data bottlenecks and enabling analysis of 500k+ vehicle cycles.
- **Conducted rigorous Hypothesis Testing** on driver behavior, identifying Depth of Discharge (**DoD**) as the primary driver of SOH degradation, while debunking correlations with speed and braking.
- **Developed an XGBoost regression model** ($MSE : 1.4$) to predict Battery SOH.
- **Influenced Product Strategy** by integrating SOH insights into the iQube mobile app, providing actionable charging recommendations to improve battery longevity and customer satisfaction.

Assistant Manager
Adani Enterprises Ltd — TC

Aug 2022 – Jan 2024
Ahmedabad, GJ

- **Automated** SAP workflows using VBA and analyzed 15 years of historical datasets to optimize vendor selection, driving 16 Lakhs in cost savings and reducing buyer turnaround time by 3 hours.

PROJECTS

Full-Stack RAG AI Agent

[Repo](#)

- Engineered a stateful AI Agent using **LangGraph** and **FastAPI**, moving beyond simple chat chains to sophisticated, multi-turn conversation flows with persistent memory.
- **Orchestration & ETL**: Implemented a custom Retrieval-Augmented Generation (**RAG**) pipeline using **FAISS** and **OpenAI Embeddings**, enabling the agent to index and query user-uploaded PDFs dynamically.
- **Model Lifecycle Management**: Integrated **OpenAI Function Calling** to create an autonomous system capable of executing external tools based on user intent.
- **Serving**: Orchestrated the entire stack using **Docker Compose** and Nginx reverse proxy, ensuring a production-ready, containerized deployment.

Real-Time Traffic MLOps Pipeline

[Repo](#)

- Designed and deployed a containerized machine learning pipeline to predict urban traffic delays, automating the lifecycle from data ingestion to model deployment.
- **Orchestration & ETL**: Developed complex Apache **Airflow** DAGs to ingest real-time data from the TomTom API (5-minute intervals) and perform daily ETL jobs converting raw logs to optimized Parquet files.
- **Model Lifecycle Management**: Integrated **MLflow** for experiment tracking and model registry, implementing an automated "Champion vs. Challenger" promotion logic to ensure only the best-performing models reach production.
- **Serving**: Deployed the production model via a high-performance **FastAPI** microservice, containerized the entire stack using Docker/Docker Compose for consistent environments.

ECU Data Pipeline: DCM to MATLAB

[Repo](#)

- **Automated ELT Pipeline**: Engineered a zero-config, portable tool utilizing a custom RegEx engine to extract complex structures from proprietary automotive files (.DCM) into Excel/MATLAB, optimizing data integrity with multi-encoding support.

EDUCATION

2018 - 2022 B.Tech **NIT Calicut** – First class with distinction
2015 - 2017 Board of Intermediate at **Aditya jr. college**

(CGPA: 8.03/10)
(Percentage: 94%)

Last updated: May 8, 2026